

AI WEBSITE DESIGN

PROJECT PRESENTATION





INTRODUCTION

Prompt Engineering for Web Design is the process of creating clear, structured, and effective instructions that guide an Artificial Intelligence (AI) system to produce useful web design results.

AI tools can help a web designer:

- Generate HTML
- Generate CSS
- Generate JavaScript
- Create website layouts
- Design user interfaces
- Generate website content
- Create images and graphics
- Improve existing designs
- Make websites responsive



INTRODUCTION

- Debug code
- Explain code
- Generate database structures
- Generate API code
- Improve SEO
- Test website usability
- Convert ideas into complete website prototypes

However, the quality of the result depends heavily on the quality of the instructions given to the AI.

This leads to an important principle:

Better prompts generally produce better and more controllable results.

Prompt engineering therefore becomes an important skill for modern web designers.

What Is a Prompt

A **prompt** is an instruction, question, command, or description given to an AI system to produce a desired result.

For example:

“Create a website for a restaurant.”

This is a prompt, but it is very general.

A more detailed prompt could be:

“Create a modern responsive restaurant website using HTML5, CSS3 and Bootstrap 5.”

What Is a Prompt

The website should include:

- Responsive navigation bar
- Hero section
- About section
- Menu section
- Customer testimonials
- Contact form
- Google Maps section
- Footer

Use a professional restaurant design with warm colors, large food images, rounded cards and clear call-to-action buttons.

Make the website responsive on mobile, tablet and desktop.
The second prompt gives the AI much more information to work with.



What Is Prompt Engineering

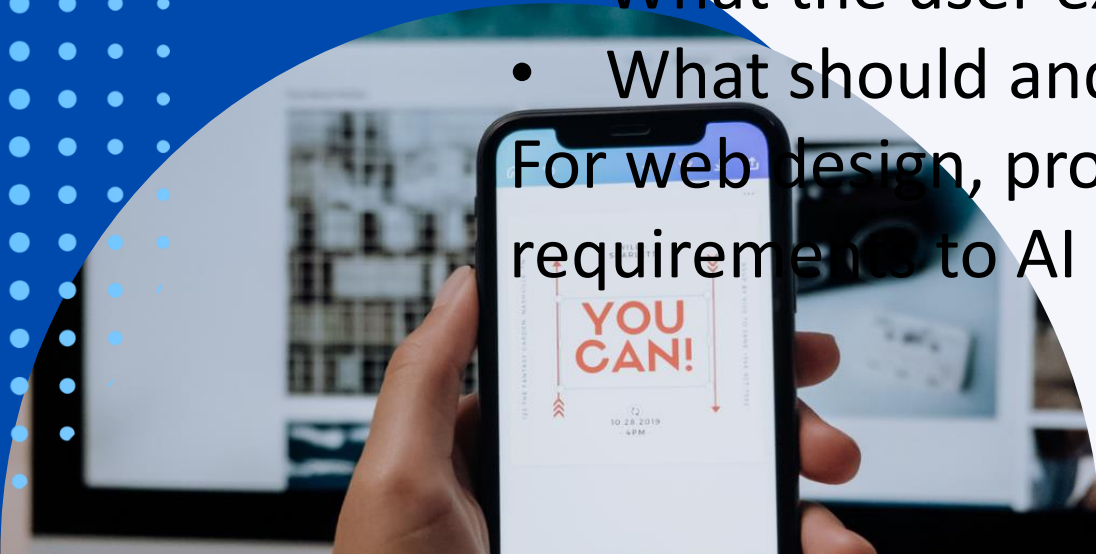
Prompt engineering is the practice of designing and refining prompts to obtain accurate, useful, consistent, and controllable AI-generated results.

It is not simply asking AI questions.

It involves understanding:

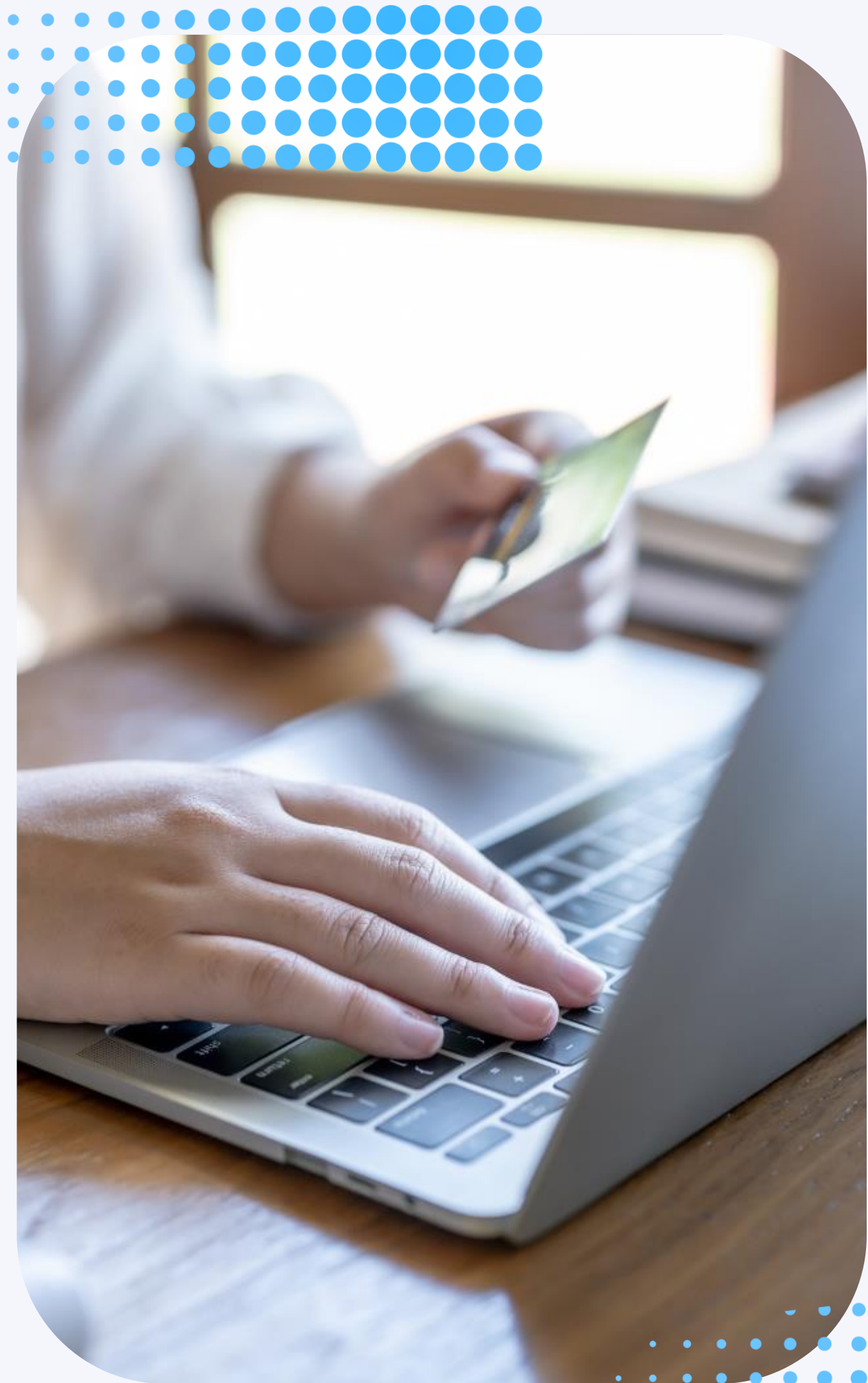
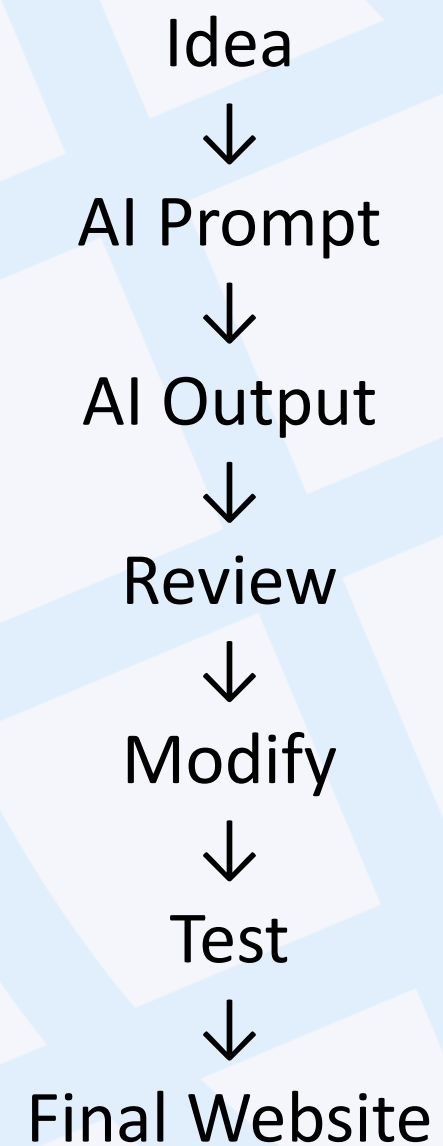
- What you want
- How to describe it
- What technology should be used
- What constraints exist
- What output format you need
- What the user experience should be
- What should and should not be included

For web design, prompt engineering allows you to communicate your design requirements to AI in a structured way.



Why Prompt Engineering Is Important in Web Design

Traditional web development requires the designer/developer to manually create many parts of a website. With AI assistance, many tasks can be accelerated. For example:



Why Prompt Engineering Is Important in Web Design

Without a good prompt:

Vague prompt



Poor output



Many corrections



Wasted time

With a good prompt:

Detailed prompt



Relevant output



Small corrections



Faster development



Prompt Engineering Does Not Replace Web Design Knowledge

This is extremely important.

AI can generate code, but a web designer should understand what the code is doing.

For example, AI may generate:

```
.container {  
  display: flex;  
  justify-content: space-between;  
}
```

The designer should understand:

What display: flex does

What justify-content does

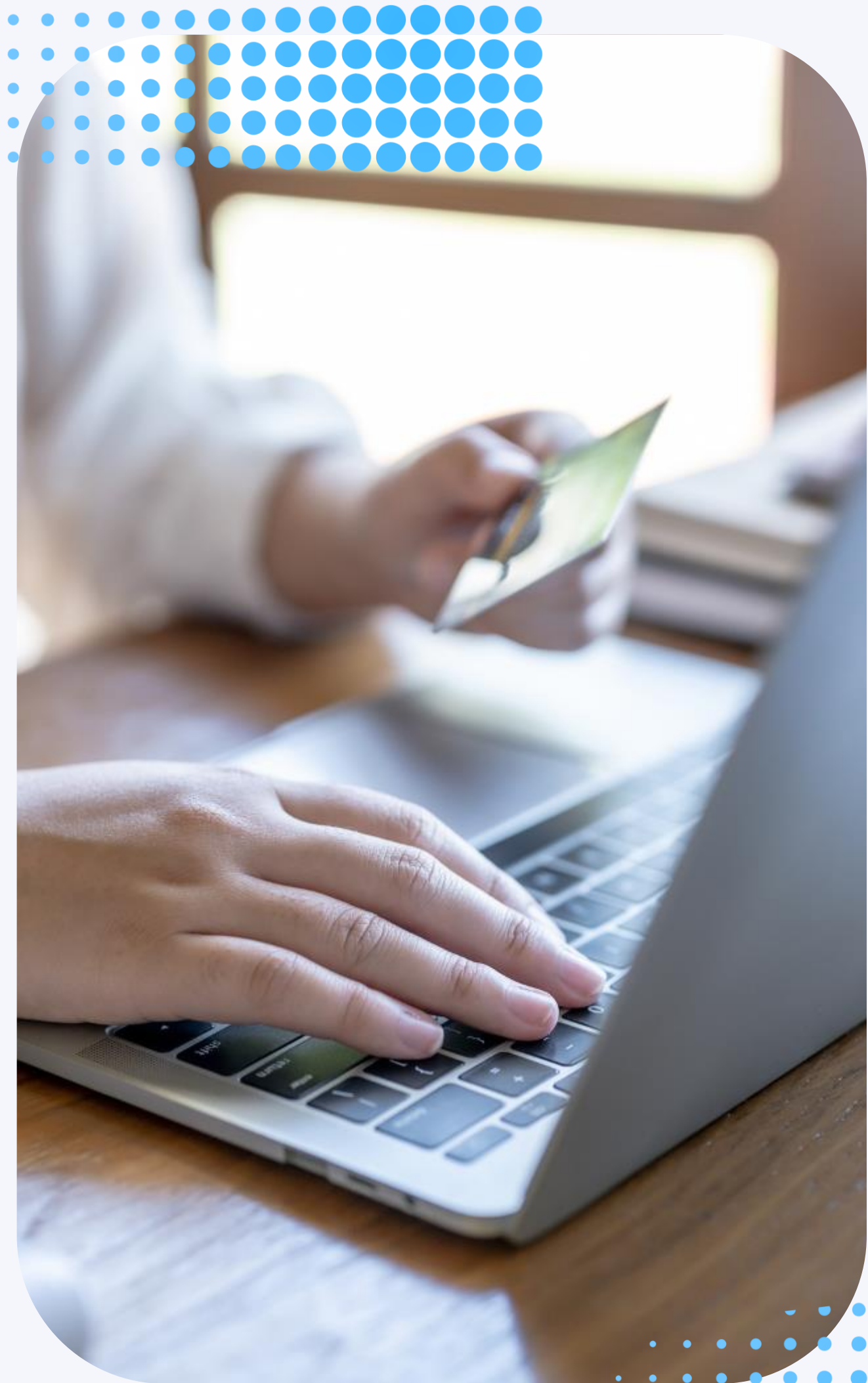
When Flexbox should be used

How it behaves on mobile

Whether the layout is appropriate

Therefore:

AI should be treated as an assistant, not as a replacement for web design knowledge.



Components of a Good Web Design Prompt

A good web design prompt can contain several important components.

A useful structure is:

ROLE

+

TASK

+

PROJECT CONTEXT

+

TECHNOLOGY

+

DESIGN REQUIREMENTS

+

FUNCTIONAL REQUIREMENTS

+

RESPONSIVENESS

+

CONSTRAINTS

+

OUTPUT FORMAT



Components of a Good Web Design Prompt

For example:

Act as a professional UI/UX designer and frontend developer.

Create a modern landing page for a digital payment platform.

Technology:

- HTML5
- CSS3
- Bootstrap 5
- JavaScript

Design:

- Modern
- Professional
- Clean
- Mobile-first
- Minimal



Components of a Good Web Design Prompt

Sections:

- Navbar
- Hero
- Services
- Features
- Testimonials
- FAQ
- CTA
- Footer

The website must be responsive on mobile, tablet and desktop.

Return the complete HTML, CSS and JavaScript code.





The Role in a Prompt

A **role** tells the AI what type of expert you want it to behave like.

For example:

Act as a professional web designer.

Or:

Act as a senior frontend developer.

Or:

Act as a UI/UX designer specializing in modern SaaS websites.

Or:

Act as a senior Bootstrap developer.



The Role in a Prompt

Why roles are useful

A role provides context.

Compare:

Create a dashboard.

with:

Act as a senior UI/UX designer and frontend developer.

Design a professional admin dashboard for an e-commerce platform.

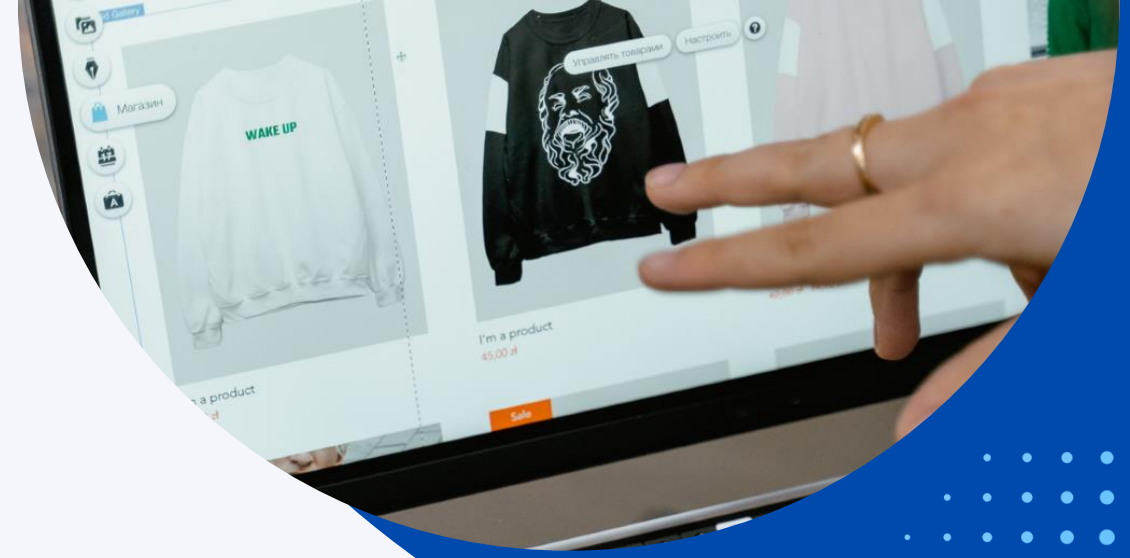
The second prompt establishes the expected expertise and context.

Defining the Task

The task tells AI exactly what you want it to accomplish.
Examples:

- Create a landing page.
- Generate the HTML structure for this design.
- Convert this design into Bootstrap 5.
- Make this website responsive.
- Improve the navigation bar.
- Debug this JavaScript code.
- Create a dashboard UI.

A task should be specific.





Providing Context

Context explains what the project is about.
For example:

“I am building a digital services platform where users can purchase airtime, data, TV subscriptions and other digital services.”

This is much better than simply saying:

“Create a dashboard.”

The AI now understands the type of application it is designing.



Specifying the Target Users

A good prompt can describe who will use the website.
For example:

“The website is designed for Nigerian university students between the ages of 18 and 30.”

Or:

“The dashboard will be used by business owners who may not have technical knowledge.”

Specifying the Target Users

This can influence:

- Navigation
- Language
- Typography
- Layout
- Button labels
- Complexity
- User experience



Specifying the Technology

Always specify the technology when it matters.
For example:

Use:

- HTML5
- CSS3
- Bootstrap 5
- Vanilla JavaScript

Or

Use:

- React
- Vite
- Tailwind CSS
- React Router



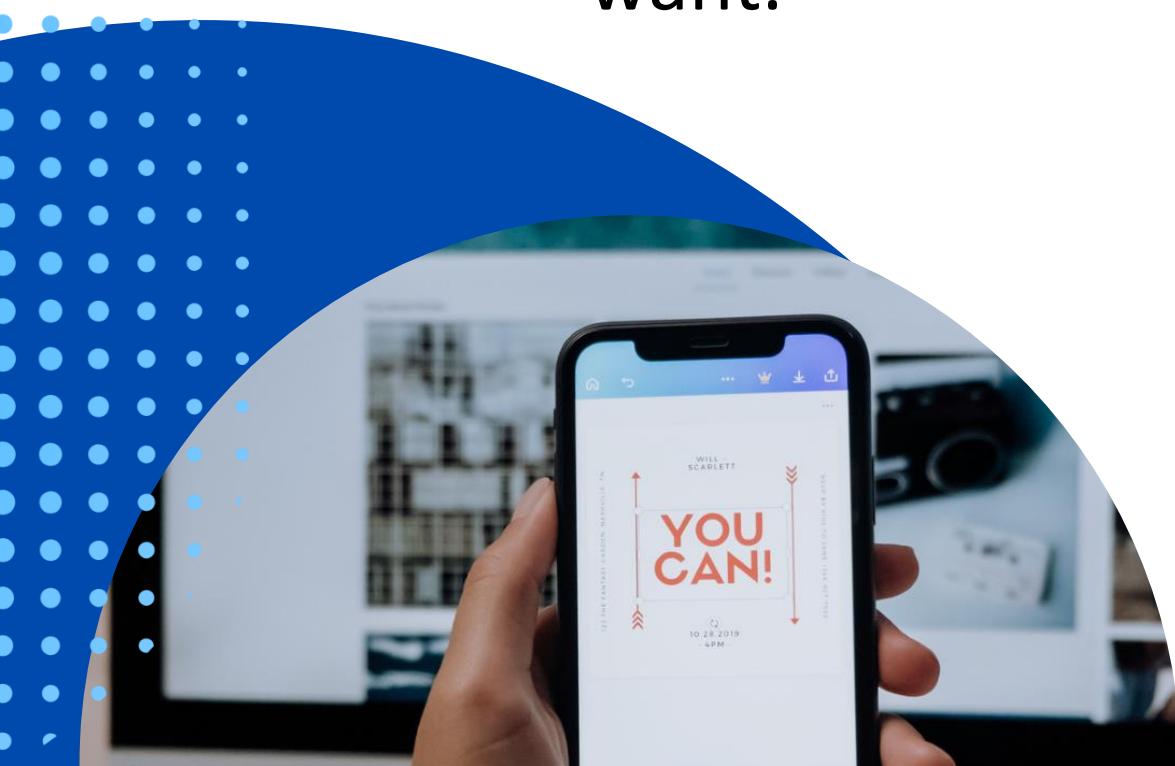
Specifying the Technology

Or:

Use:

- PHP
- MySQL
- Bootstrap
- JavaScript

Without specifying the technology, AI may choose a framework you don't want.



Specifying the Design Style

Describe the visual style you want.

Examples:

Modern

Professional

Minimal

Elegant

Corporate

Luxury

Playful

Creative

Futuristic

Clean

Tech-focused

You can combine styles:

Create a modern, clean and professional fintech interface.



Specifying Colors

Colors can be included in the prompt.

For example:

Use a blue and white color scheme with dark gray text.

Or more specifically:

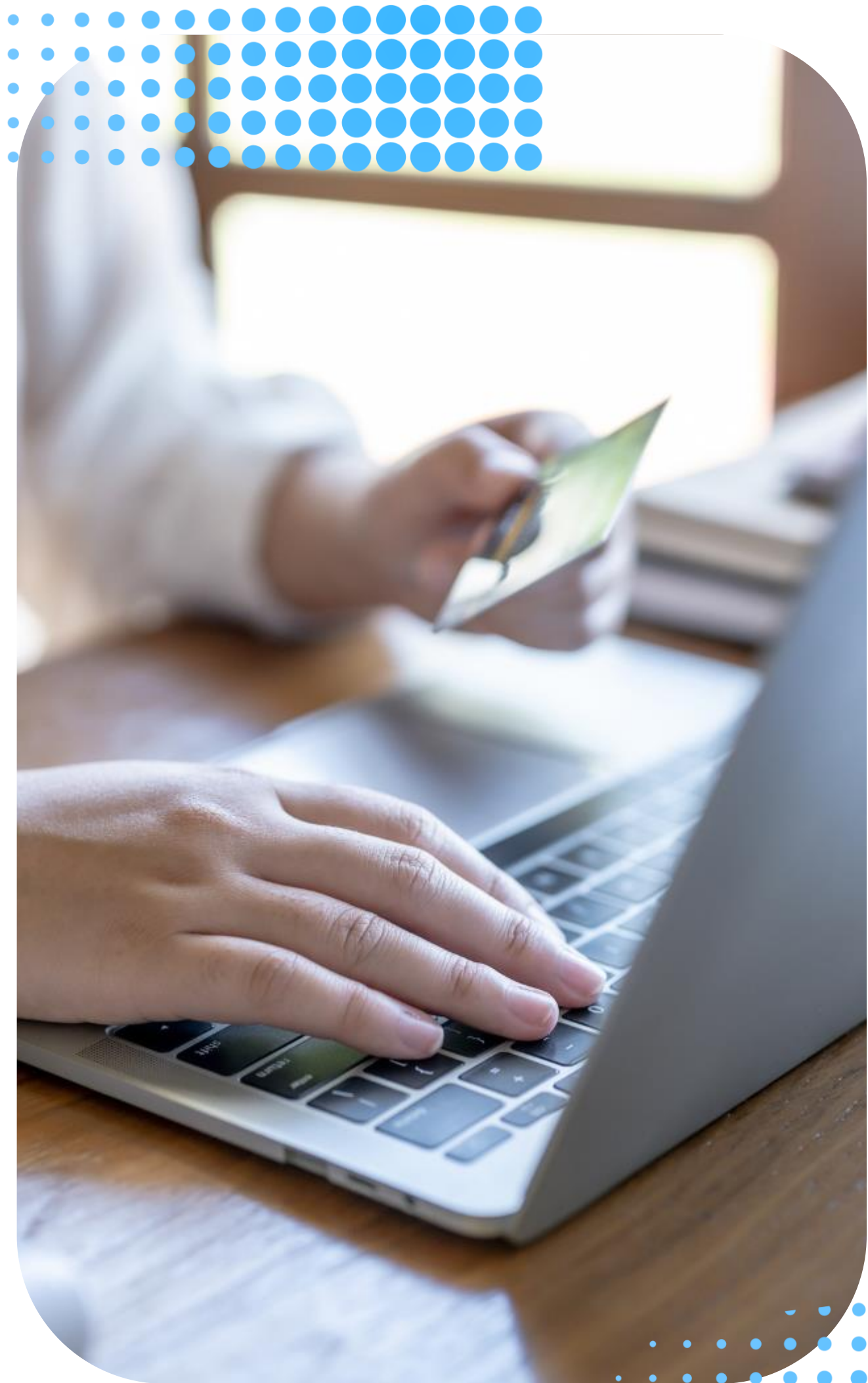
Primary color: #2563EB

Secondary color: #0F172A

Background: #F8FAFC

Text: #1E293B

Specific color requirements give AI more control over the final design.



Specifying Typography

You can tell AI what type of typography you want.

Example:

Use a clean sans-serif font.

Use:

- Large bold headings
- Medium-sized subheadings
- Comfortable body text
- Clear line spacing

You can also specify a font:

Use Inter as the primary font.



Describe how the website should be arranged.

For example:

Create a two-column hero section.

Left side:

- Heading
- Description
- CTA buttons

Right side:

- Product illustration
- Floating feature cards

Instead of:

Create a hero section.

The detailed version gives AI a clearer layout.

Specifying Layout



Specifying Website Sections

Tell AI exactly what sections you want.

For example:

Include:

1. Navigation
2. Hero
3. Services
4. Features
5. How It Works
6. Testimonials
7. Pricing
8. FAQ
9. Call-to-action
10. Footer

This reduces the chance that important sections will be forgotten.





Specifying Components

You can ask AI to create specific UI components.

Examples:

Create a responsive navbar.

Create a product card.

Create a pricing card.

Create a login form.

Create a responsive admin sidebar.

Create a modal for confirming transactions.

Specifying User Interactions

A web design prompt should describe how users should interact with the interface.

For example:

“When the user clicks the hamburger icon, open the mobile navigation menu.

When the menu is open, clicking outside the menu should close it.”

Another example:

“When the user submits the form, validate all required fields and display appropriate error messages below the fields.”

This moves the prompt from static design into interactive design.



Specifying Responsive Behavior

Don't simply say:
“Make it responsive.”

Be more specific.

For example:

Desktop:

- Four product cards per row.

Tablet:

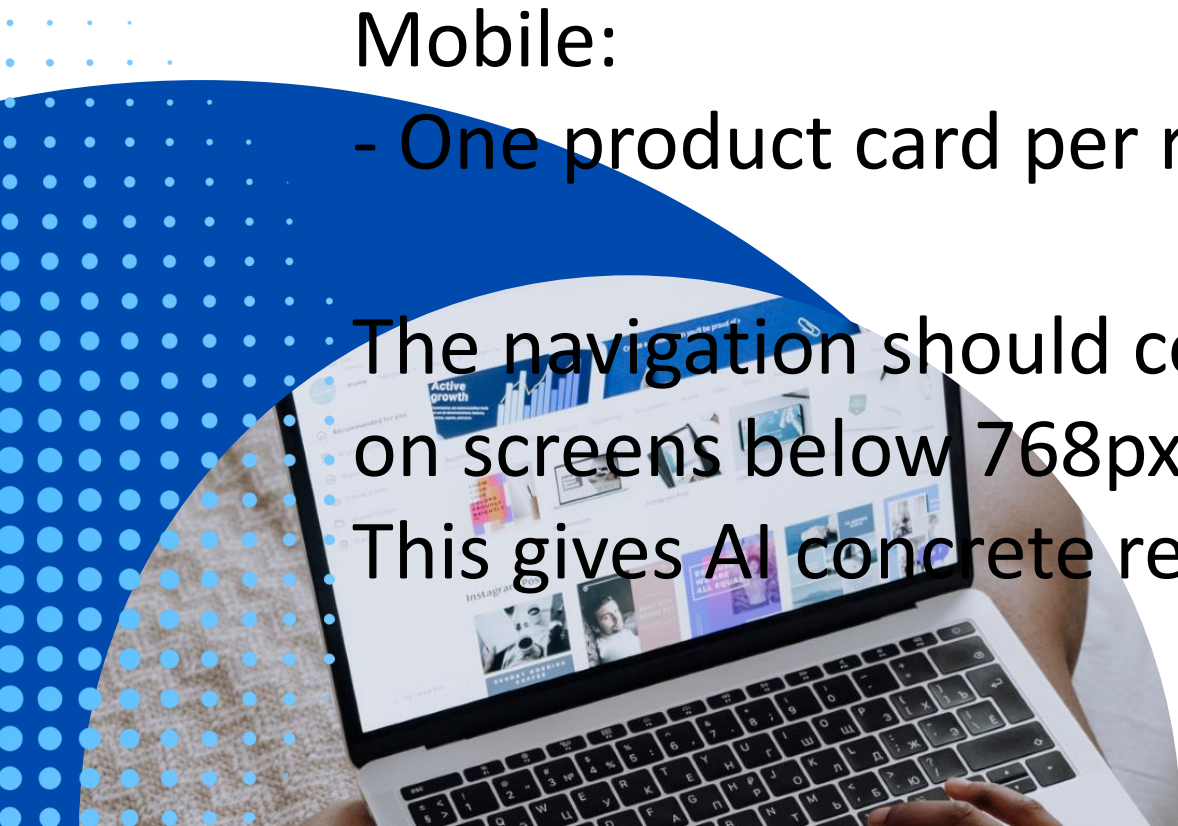
- Two product cards per row.

Mobile:

- One product card per row.

The navigation should collapse into a hamburger menu
on screens below 768px.

This gives AI concrete responsive rules.





Specifying Accessibility

Accessibility can also be included in your prompt.

Example:

Follow basic web accessibility best practices.

Requirements:

- Use semantic HTML
- Add alt text to images
- Use labels for form inputs
- Ensure buttons are keyboard accessible
- Maintain sufficient color contrast
- Use meaningful link text

This encourages AI to produce more accessible websites.



Specifying SEO Requirements

You can ask AI to consider SEO.

Example:

Optimize the page structure for SEO.

Requirements:

- Use one H1
- Use logical H2 and H3 headings
- Add a descriptive title
- Add a meta description
- Use semantic HTML
- Add descriptive alt text



Specifying Performance Requirements

You can also tell AI to optimize performance.

Example:

Optimize the website for performance.

Requirements:

- Avoid unnecessary JavaScript
- Use responsive images
- Lazy-load images where appropriate
- Avoid excessive animations
- Keep CSS organized
- Avoid unnecessary external libraries

Specifying Code Structure

When asking AI to generate code, specify how you want it organized.

For example:

Separate the project into:

index.html
css/style.css
js/script.js
images/

Or:

Create separate React components for:

Navbar
Hero
Services
Testimonials
Footer

This makes the generated project easier to maintain.



Specifying Output Format

AI needs to know how you want the result presented.

For example:

Return the result as:

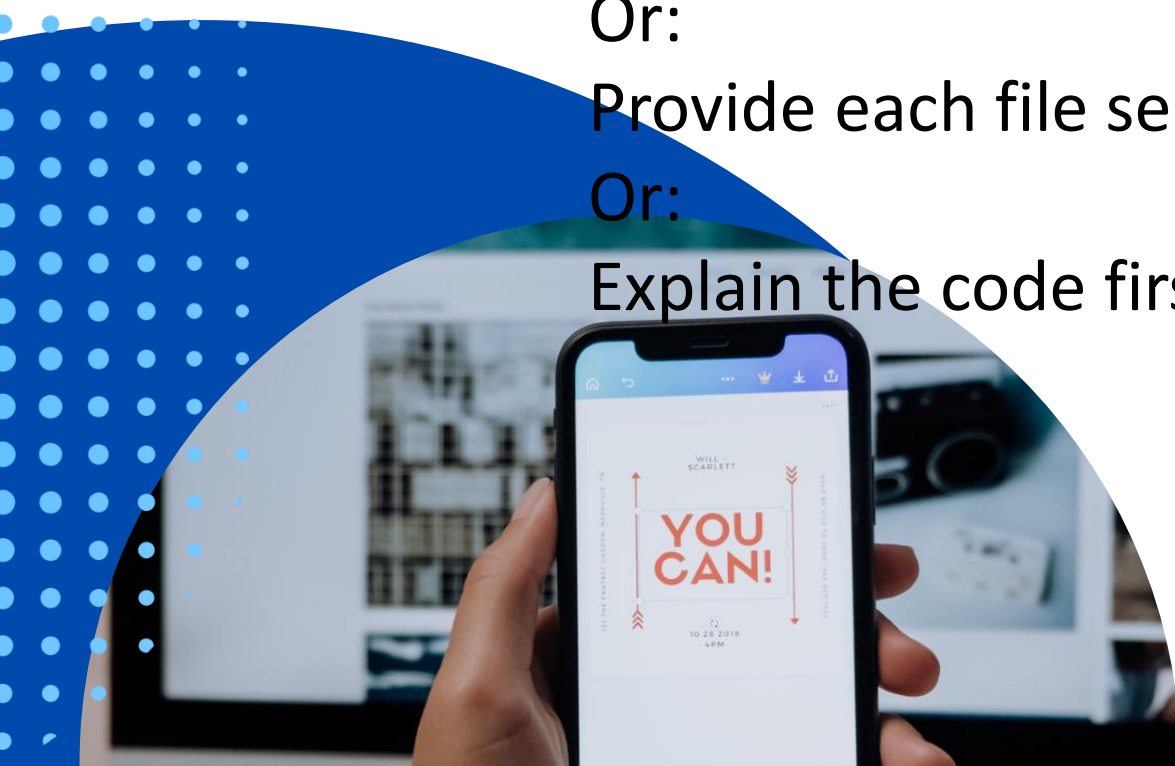
1. HTML
2. CSS
3. JavaScript

Or:

Provide each file separately using code blocks.

Or:

Explain the code first, then provide the complete code.



The background features a light blue globe with a grid of latitude and longitude lines, centered behind the text. Surrounding the globe are several large, solid blue circles of varying sizes. Some of these circles have a pattern of smaller, lighter blue dots on their edges, creating a halftone or dotted effect. The overall design is clean and modern, with a focus on geometric shapes and a monochromatic blue color palette.

**THANK
YOU**